

COMP111: Artificial Intelligence

Views of AI and History of AI

Frank Wolter

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Learning outcomes

Give and criticise definitions of AI; distinguish narrow AI from AGI; explain the four views of AI; explain the combinatorial explosion and why it derailed early AI; name the key milestones from 1943 to today and the technical reason each one worked.

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- ▶ Diagnosis in healthcare; drug discovery; protein structure prediction.
- ▶ Recommendation, fraud detection, forecasting: the “invisible” AI you use every day.

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Vote

Which of these is AI? (a) a chess engine searching millions of positions; (b) a spam filter trained on a million emails; (c) a thermostat; (d) a sat-nav running Dijkstra; (e) ChatGPT writing a poem.

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Compare aeronautics: understanding *how birds fly* is a different project from *building an aircraft*. Aircraft do not flap their wings — but they do obey aerodynamics.

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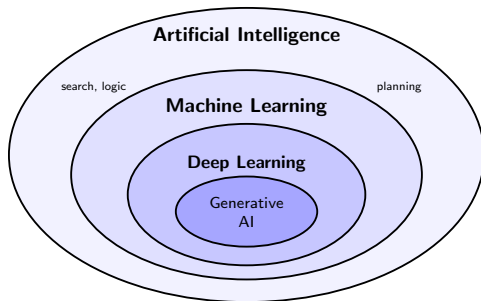
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Before ChatGPT, other large language models (LLMs), and more generally generative AI, came up in 2022/23, almost all AI researchers and practitioners would have described their goal as narrow AI. That has changed a bit: some argue that generative AI is a significant step towards AGI, while others regard current systems as very broad *narrow* AI.

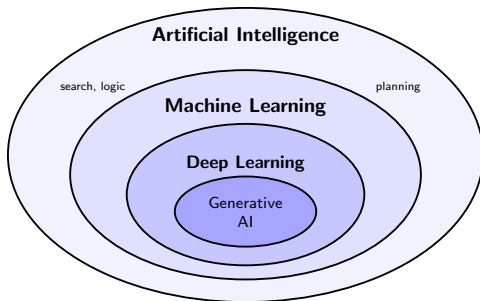
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- ▶ **Machine learning:** behaviour is *learned from data* rather than programmed.
- ▶ **Deep learning:** ML with many-layered neural networks; features are learned, not hand-crafted.
- ▶ **Generative AI:** models that *generate* content (text, images, code) from a prompt.
- ▶ Not all AI is ML: a chess engine's search, or a Prolog program, learns nothing.

Four Views of AI

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Modern AI lives mostly in the **bottom-right cell**: agents that act rationally to achieve a goal.

Thinking Humanly vs Acting Humanly

- ▶ Cognitive science: to understand and build intelligent machines we should simulate human thinking.
- ▶ Computer science (often): to build intelligent machines simulate the intelligent behavior of humans. It is not necessary nor desirable that an intelligent machine processes information in the same way as humans (think humanly); the aim is to create machines that are able to behave (e.g., take part in conversations) in the same way as humans.

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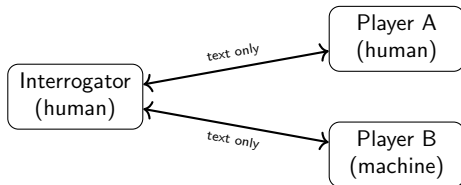
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“Can machines think?” → “Can machines behave intelligently?”

The Turing Test (1950)

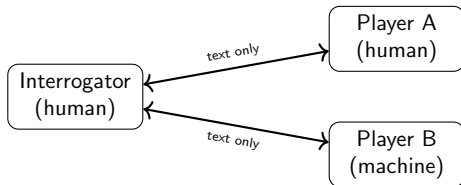
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30 seconds with your neighbour

Write down *one* question you think a machine would fail but a human would pass. (Classics: commonsense physics; ambiguous pronouns — “the trophy doesn’t fit in the suitcase because *it* is too big”.)

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Modern practice: replace the Turing test by **task-specific benchmarks** — and remember Goodhart's law (“when a measure becomes a target, it ceases to be a good measure”).

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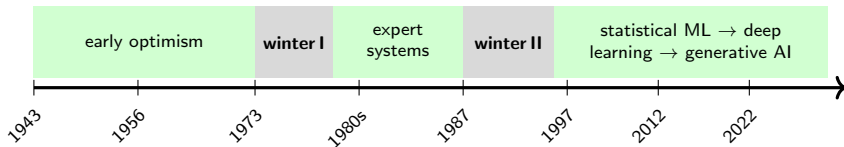
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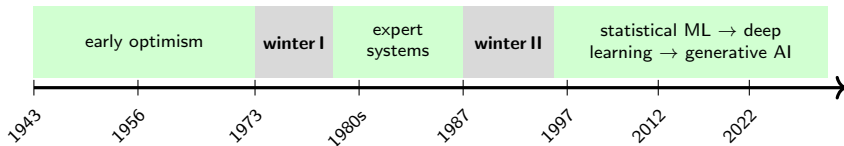
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- ▶ Rationality is not omniscience: doing the best you can with the information and computation available is called **bounded rationality**.
- ▶ Problem of current generative AI: it often does not reason correctly.

A brief history of AI

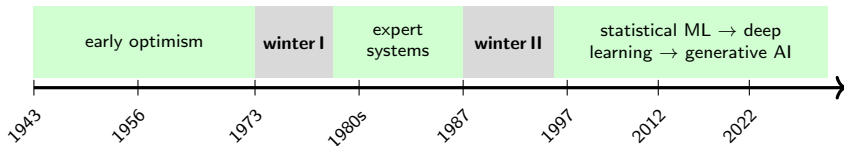


A brief history of AI



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For every milestone that follows, ask: *what exactly made it work, and what did it not solve?*

First steps towards AI: 1943–56

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- ▶ Minsky and Edmonds (1951) build the first neural network computer (SNARC) using vacuum tubes to simulate a network of 40 neurons.
- ▶ Shannon, Turing (1950): the first chess playing programs. (Turing's had to be executed by hand — no machine was fast enough.) Turing also proposed building a “child machine” and *educating* it: learning was part of the plan from the very beginning.

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- ▶ The corresponding article was rejected by the *Journal of Symbolic Logic*.
- ▶ The name ‘Artificial Intelligence’, coined by John McCarthy, was adopted.
- ▶ No technical breakthroughs — but a research agenda, a community, and a name.

Optimism regarding AGI (1956–70)

Promising performance of early AI systems on simple examples (integration problems, algebra problems, moving blocks) led to great expectations. Simon (1957):

It is not my aim to surprise or shock you — but the simplest way I can summarize is to say that there are now in the world machines that can think, that can learn and that can create. Moreover, their ability to do these things is going to increase rapidly until — in a visible future — the range of problems they can handle will be coextensive with the range to which the human mind has been applied.

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Also from this era: Rosenblatt's **perceptron** (1958), an early trainable neural network, with huge press attention. Minsky & Papert (1969) then showed that a single-layer perceptron cannot even learn XOR — and neural network funding dried up for fifteen years.

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Estimate first

In chess, how many distinct positions can arise after *four* moves by each player? (a) thousands (b) millions (c) billions (d) hundreds of billions

Combinatorial Explosion (Chess)

In principle, it is easy to write a program playing chess by computing all possible sequences of moves (say, up to 40 moves) and then selecting an optimal move. In chess, however, there are approximately

- ▶ 400 different positions after one move apiece;
- ▶ 72,084 different positions after two moves apiece;
- ▶ 9×10^6 positions after three moves apiece;
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In practice, it simply is not possible (and never will be possible) to compute all relevant sequences of moves! Growth is **exponential** in the depth: b^d for branching factor b .

Combinatorial Explosion (Rubik's Cube)

For a Rubik's Cube there are

- ▶ 43,252,003,274,489,856,000 combinations ($\approx 4.3 \times 10^{19}$);
- ▶ up to 481,229,803,398,374,426,442,198,455,156,736 brute-force solution attempts.

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Insight: the fact that a program can find a solution *in principle* does not mean it can find it *in practice*. And yet every cube can be solved in **at most 20 moves** (“God’s number”, proved 2010): good **heuristics** explore a vanishing fraction of the space. (Topic of the next section.)

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Insight: the fact that a program can find a solution *in principle* does not mean it can find it *in practice*. And yet every cube can be solved in **at most 20 moves** (“God’s number”, proved 2010): good **heuristics** explore a vanishing fraction of the space. (Topic of the next section.)

The **Lighthill Report** (1973) for the British Research Council concluded that AI researchers had failed to address the combinatorial explosion, and was pessimistic about general robotics and natural language processing. It formed the basis for the UK government’s decision to essentially end support for AI research: the first **AI winter**.

Expert Systems (Narrow AI)

- ▶ General purpose, brute force techniques don't work, so use **knowledge rich** solutions. The early 1980s saw the emergence of **expert systems**: systems exploiting knowledge about tightly focused domains to solve problems normally considered the domain of experts. Architecture: a **knowledge base** of IF-THEN rules elicited from experts, plus an **inference engine**.

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 - ▶ lack of **trust** in the recommendations given.
- ▶ Result: the second **AI winter** ($\approx 1987-93$). The field then turned quietly from hand-coded rules to **learning from data** and to **probability** (Bayesian networks, statistical NLP).

Major steps in the past 30 years

1997: IBM's Deep Blue defeated world chess champion Garry Kasparov ($3\frac{1}{2} : 2\frac{1}{2}$).

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Note: *no learning at all*. Deep Blue is a triumph of search and engineering.



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- ▶ Reality check: the follow-up “Watson Health” business was sold off in 2022. Winning a quiz show is not practising medicine.

AlphaGo wins against a 9-dan professional (2016)

- ▶ Go: two players place stones on the intersections of a 19×19 grid; the objective is to surround more territory than the opponent. About 2×10^{170} legal positions, and positions are hard to evaluate with hand-written rules.
- ▶ AlphaGo (Google DeepMind) beat Lee Sedol 4–1.
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- ▶ AlphaZero (2017): given only the rules and no human games, superhuman at Go, chess and shogi by self-play.



AlphaFold: predicting 3D protein structures (2020)

- ▶ Proteins underpin the biological processes of all living beings. They are the building blocks of life.
- ▶ There are hundreds of millions of known proteins. Each has a unique 3D shape that determines how it works and what it does.
- ▶ Predicting that 3D shape from the sequence of amino acids had been an unsolved challenge for over 50 years; determining one structure experimentally could take years.
- ▶ Essentially solved by Google DeepMind in 2020; the public database now covers over 200 million proteins. Nobel Prize in Chemistry 2024 for Demis Hassabis and John Jumper.
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The pattern: a hard scientific problem + a large, clean dataset + deep learning.

Self-driving cars and RoboCup

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- ▶ Why is football harder than chess? **Moravec's paradox**: high-level reasoning needs little computation; low-level perception and motion need enormous amounts.

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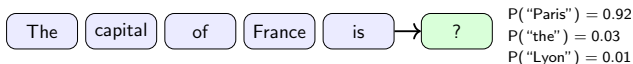
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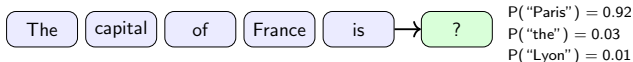
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- ▶ **Post-training** matters as much as pre-training: instruction tuning and reinforcement learning from human feedback (RLHF) turn a text predictor into an assistant.

How an LLM works — and how it fails



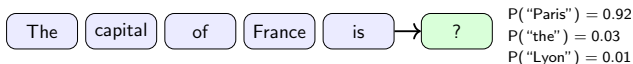
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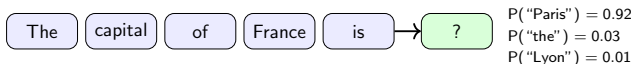
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- ▶ **Agents** chain these together: read a repository, edit code, run the tests, iterate. Note that this reunites machine learning with the old traditions of search and knowledge representation.

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- ▶ Warning: an explanation that is *plausible* but not *faithful* is worse than none.

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- ▶ Two AI winters were caused by **over-promising** plus the **combinatorial explosion**: what works on toy problems need not scale.
- ▶ Each milestone has a technical reason: Deep Blue (search + hardware), Watson (NLP + knowledge extraction), AlphaGo (search + deep RL), AlphaFold (deep learning + data), LLMs (transformers + scale + post-training).